One sky, many instruments: general formulation and approximation ladder

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Purpose. This note starts from the most general measurement equation. No form is assumed for the noise, the background or the instrument response.

It then lists, one at a time, the approximations that can be made. For each it states what the approximation breaks or creates in the way of degeneracies, and what the current implementation (the `fieldop` package) does. The aim is to choose the functional forms deliberately, for an outside deep-thinking review.

0. Notation

Symbol	Meaning
e	exposure, i.e. one detector image; $d(e)$ its detector, $f(e)$ its filter, $[t_e, t_e + \Delta t_e]$ its time span
$\mathbf{x}$	pixel position on the detector (pixels); $\tilde{\mathbf{x}} \in [-1, 1]^2$ normalised over the detector
$\mathbf{p} = F_d(\mathbf{x})$	position in the telescope focal plane (e.g. JWST $V2, V3$)
θ	position on the sky (tangent plane of a sky patch, arcsec)
$\theta_e(\mathbf{x}, \lambda)$	sky position imaged at pixel $\mathbf{x}$ (the geometric mapping)
$\mathbf{u} = \theta - \theta_e(\mathbf{x}, \lambda)$	sky offset from that position
$\lambda,\ t$	wavelength, time
$I(\theta, \lambda, t)$	astronomical sky (specific intensity)
$I_{\text{fg}}(\theta, \lambda, t)$	foreground radiance: zodiacal light, Galactic cirrus, stray light from sources outside the field
$E_e(\mathbf{x}, \lambda, t)$	instrument emission reaching the pixel without passing through the sky: thermal self-emission, glow
$D_e(\mathbf{x}, t)$	dark current
$d_e(\mathbf{x})$	recorded value

1. General measurement equation

1.1 Expected signal (no approximation)

The charge collected in pixel $\mathbf{x}$ during exposure e is a functional of everything that reaches it. The most general form linear in incoming radiance, followed by a general detector map, is

$$Q_e(\mathbf{x}) = \int_{t_e}^{t_e+\Delta t_e} dt \int d\lambda \left[\int d^2\theta \; \mathcal{R}_e(\mathbf{x}, \theta, \lambda, t) \, (I + I_{\text{fg}})(\theta, \lambda, t) \; + \; E_e(\mathbf{x}, \lambda, t) \right] + \int dt \; D_e(\mathbf{x}, t),$$

$$\mu_e(\mathbf{x}) \; = \; \mathcal{D}_e\big[\, Q_e(\cdot), \; \mathcal{H}_e \,\big](\mathbf{x}).$$

- **Response.** $\mathcal{R}_e(\mathbf{x}, \theta, \lambda, t)$ is the full response of pixel $\mathbf{x}$ to radiance from direction θ at wavelength λ and time t. It covers geometry, optics, filter transmission, quantum efficiency, intra-pixel response and charge diffusion. It is not assumed local, separable or shift-invariant.
- **Detector map.** $\mathcal{D}_e$ is non-linear and non-local, acting on the whole charge image.
  - It includes non-linearity, inter-pixel capacitance, brighter-fatter and saturation.
  - It includes ramp fitting and the calibration steps that turn ramps into `_rate` or `_cal` products.
  - $\mathcal{H}_e$ is the detector history: persistence from earlier exposures.
- **Background.** Background is not a separate additive term here. Foreground radiance passes through the same response as the sky; instrument emission and dark current enter separately.

1.2 Likelihood (no approximation)

The data are a random draw given the expected signal and the detector state:

$$d_e \; \sim \; p(d_e \,|\, \mu_e, \; \mathcal{H}_e).$$

In general this includes:

- Poisson statistics of the collected charge;
- read noise;
- correlated noise (1/f, appearing as row and column structure; inter-pixel capacitance correlating neighbours);
- cosmic-ray hits and other outliers;
- the non-Gaussian, data-dependent errors introduced by ramp fitting.

The noise is neither additive nor Gaussian in general.

1.3 What is estimated

The sky I is estimated jointly with every nuisance: $\mathcal{R}_e$, I_{fg}, E_e, D_e, $\mathcal{D}_e$ and the noise law. Everything that is not I is the **instrument operator** of exposure e.

2. Gauges (exact degeneracies of the general equation)

These transformations leave every expected signal μ_e unchanged, so no data can fix them. They must be declared and pinned by one condition each.

Gauge	Transformation	Comment
Flux scale per λ	$I \to c(\lambda)\, I, \;\; I_{\text{fg}} \to c(\lambda)\, I_{\text{fg}}, \;\; \mathcal{R}_e \to \mathcal{R}_e / c(\lambda)$	pinned by one absolute calibration
Common blur	$I \to I * b, \;\; \mathcal{R}_e(\mathbf{x}, \cdot) \to \mathcal{R}_e(\mathbf{x}, \cdot) * b^{-1}$	the sky is defined only up to a common kernel
Common shift	$I(\theta) \to I(\theta + \mathbf{a}), \;\; \mathcal{R}_e(\mathbf{x}, \theta) \to \mathcal{R}_e(\mathbf{x}, \theta + \mathbf{a})$	pinned by one astrometric reference

Gauge	Transformation	Comment
Sky against foreground	$I \to I + J, \;\; I_{\text{fg}} \to I_{\text{fg}} - J$	exact if both are static and pass through the same response

The last gauge is broken only by structure that differs between the two components: foreground that varies with time or pointing (zodiacal light changes with season and elongation), or stray light that depends on what lies outside the field.

3. Approximation ladder

Each step is one approximation. For each: the equation, why it might help identifiability, its cost, and what `fieldop` does today.

A1. Static astronomical sky

$$I(\theta, \lambda, t) = I(\theta, \lambda).$$

- **Breaks:** a time-varying response against a static sky. Any change between epochs must be instrumental. This also breaks the sky–foreground gauge when I_{fg} varies.
- **Cost:** variable sources, transients and proper motions are absorbed elsewhere. They could be added back as a sparse time-varying component.
- **fieldop:** assumed.

A2. No change within an exposure

The response, foreground and emission are constant over $[t_e, t_e + \Delta t_e]$:

$$\int dt\, \mathcal{R}_e(\mathbf{x}, \theta, \lambda, t) \to \Delta t_e\, \mathcal{R}_e(\mathbf{x}, \theta, \lambda).$$

- **Cost:** jitter and drift within an exposure are absorbed into the effective kernel; time-variable foreground is averaged.
- **fieldop:** assumed.

A3. Localised response: geometry ∘ kernel ∘ amplitude

$$\mathcal{R}_e(\mathbf{x}, \theta, \lambda) \;=\; A_e(\mathbf{x}, \lambda)\; k_e(\theta - \theta_e(\mathbf{x}, \lambda);\; \mathbf{x}, \lambda), \qquad \int k_e(\mathbf{u}; \mathbf{x}, \lambda)\, d^2u = 1.$$

This is a reparametrisation of any response with a finite first moment, not a restriction. It splits $\mathcal{R}_e$ into three parts:

- **Geometry** $\theta_e(\mathbf{x}, \lambda)$: where the pixel looks.
- **Shape** k_e: how the light is spread around that point.
- **Amplitude** A_e: how much light is collected.

It introduces a new **internal gauge:** the first moment of k_e trades against θ_e. It is pinned by fixing the centroid of the core of k_e to zero.

Geometry.

$$\theta_e(\mathbf{x}, \lambda) \;=\; W_e\Big(\mathbf{x} + \Delta_{\text{fp}}(F_d(\mathbf{x}), \lambda, t_e) + \Delta_d(\mathbf{x})\Big).$$

- W_e is the pointing: the header WCS, refined by a registration shift and a per-exposure affine.
- Δ_{fp} is optical distortion, continuous over the focal plane and possibly chromatic and slowly varying.
- Δ_d is a per-detector term: mounting and pixel-grid irregularities.

Shape. The shape splits naturally into a part continuous over the focal plane (optics) and a per-detector part (pixel response, charge diffusion):

$$k_e(\mathbf{u};\mathbf{x},\lambda) \;\simeq\; \big[\, k^{\text{opt}}(\,\cdot\,;\mathbf{p},\lambda,t_e) * k_d^{\text{det}}(\,\cdot\,;\mathbf{x},\lambda)\,\big](\mathbf{u}), \qquad \mathbf{p} = F_d(\mathbf{x}).$$

- **Breaks:** continuity of the optics across detector boundaries ties neighbouring detectors together. The detector part carries discontinuities.
- **fieldop:** uses one shape field conditioned on (detector embedding, $\tilde{\mathbf{x}}$, λ, epoch, exposure code), with no focal-plane coordinate. Optical continuity across detectors is therefore not used.

Amplitude. It splits into the projected pixel area and the throughput:

$$A_e(\mathbf{x},\lambda) \;=\; \Big|\det \frac{\partial \theta_e}{\partial \mathbf{x}}\Big| \;\underbrace{T_f(\lambda;\mathbf{p})\;\eta_d(\mathbf{x},\lambda)\;g_e}_{\text{filter}\,\times\,\text{QE}\,\times\,\text{gain}}.$$

- **Breaks:** the pixel area is tied to the geometry, so photometry and astrometry constrain each other.
- **Breaks:** filter transmission (smooth over the focal plane) is separable from quantum efficiency and flat structure (per detector).
- **fieldop:** uses an area term (1 for flat-fielded input), a published filter curve as input, and one learned log-throughput field over (detector, $\tilde{\mathbf{x}}$, λ, epoch) plus a per-exposure gain.

A4. Foreground through the same response

Insert A3 into §1.1 and use the fact that the foreground is smooth on the scale of the kernel, $k_e * I_{\text{fg}} \simeq I_{\text{fg}}$:

$$Q_e(\mathbf{x}) \simeq \Delta t_e \int d\lambda\, A_e(\mathbf{x},\lambda)\Big[\,(k_e \star I)(\theta_e(\mathbf{x},\lambda),\lambda) + I_{\text{fg}}(\theta_e(\mathbf{x},\lambda),\lambda,t_e)\Big] + \Delta t_e \int d\lambda\, E_e(\mathbf{x},\lambda) + \Delta t_e\, \bar{D}_e(\mathbf{x}).$$

- **Breaks:** the foreground is imprinted through the *same* amplitude A_e as the sky. It acts as a "sky flat", which ties the flat field to the background and separates throughput from additive background.
  - Additive terms not multiplied by A_e (instrument emission, dark current, stray light from outside the field) remain separate.
- **Cost:** a model for I_{fg} is needed: smooth in θ and λ, varying with epoch and pointing.
- **fieldop:** does **not** do this. The background is an additive surface $B_e(\tilde{\mathbf{x}})$ with no wavelength dependence and no multiplication by throughput (quadratic per exposure, plus an 8×8 pattern per detector, plus row/column stripes). This is the "integrated over wavelength" approximation.

A5. Linear detector, separated

$$\mathcal{D}_e[Q](\mathbf{x}) \;\simeq\; \Big(\mathcal{C}_d * \big(Q + \mathcal{B}[Q]\big)\Big)(\mathbf{x}) \;+\; \mathcal{P}_e(\mathbf{x}) \;+\; \text{(non-linearity, saturation mask)}.$$

- $\mathcal{C}_d$ is inter-pixel capacitance, a small fixed kernel per detector.
- $\mathcal{B}$ is brighter-fatter, a charge-dependent redistribution.
- $\mathcal{P}_e$ is persistence from the detector history $\mathcal{H}_e$.
- **Breaks:** inter-pixel capacitance and brighter-fatter act *after* the optics and depend on charge. They are distinguishable from the kernel by their dependence on flux.

- **Cost:** the extra terms are only partly constrained by data.
- **fieldop:** none of these are modelled. Inter-pixel capacitance and charge diffusion are absorbed into the shape; there is an optional gated non-linearity, untested; saturated pixels are masked.

A6. Wavelength quadrature

$$\int d\lambda\,(\cdots) \;\simeq\; \sum_{q=1}^{Q} w_q\,(\cdots)\big|_{\lambda_q}.$$

- **fieldop:** $Q = 5$ nodes at the quantiles of $T_f(\lambda)/\lambda$, with error 1.2×10^{-3}. Chromatic kernels and geometry are kept at each node.

A7. Noise model

There are several levels of approximation to $p(d_e \mid \mu_e)$:

Level	Likelihood	Comment
L0	exact ramp likelihood (Poisson + read noise + jumps)	needs the ramps (`_uncal`)
L1	$d \sim \mathcal{N}(\mu,\ \sigma_r^2 + \mu/g + (f\mu)^2)$ + correlated	variance from the **model**, not the data; avoids flux bias
L2	$d \sim \mathcal{N}(\mu,\ (\nu\,\sigma^{\text{pipe}})^2 + (f\mu)^2)$	pipeline error from the data, times a factor
L3	L2 with a robust loss (Huber) for outliers	**fieldop today**

Correlated noise can be modelled as covariance, or as deterministic nuisances (row/column offsets). fieldop uses deterministic stripes.

- **Breaks:** at L1, the variance depends on the model's flux. Bright sources then constrain the gain g and the throughput together, and model-based variance removes the bias of data-based errors (faint pixels that fluctuate low get small errors and too much weight).
- **fieldop measurements:** the pipeline σ is about 29% too small, and the fractional floor is about 5%.

A8. Sky representation

$I(\theta, \lambda)$ must be represented by a finite model.

- **fieldop:** 12 spectral bumps in $\log\lambda$ times a coefficient grid at 0.04″ with bilinear interpolation, one grid per sky patch:

$$I(\theta,\lambda) = \sum_{j=1}^{12} \phi_j(\log\lambda) \sum_{m,n} a_{j,mn}\,\Lambda\!\Big(\tfrac{\theta_1-\theta_{1,m}}{p_s}\Big)\Lambda\!\Big(\tfrac{\theta_2-\theta_{2,n}}{p_s}\Big).$$

- **Known limitation:** this cannot represent short-wavelength point sources. The worst residual relative to the peak is 0.26 in F090W and 0.16 in F150W, against 5×10^{-6} in F444W.
- Because of it, fieldop pins the blur gauge with a **floor:** no kernel may be sharper than one sky tent, enforced by dilating all kernels. A continuous sky representation needs a different declaration of this gauge.

4. Current implementation in one equation

With A1–A3, A6, L3 noise, the additive background and the grid sky:

$$d_e(\mathbf{x}) \;\approx\; \sum_{q=1}^{Q} w_q\, \alpha_e(\mathbf{x})\, g_e(\mathbf{x}, \lambda_q) \int d^2u\; k_e(\mathbf{u}; \mathbf{x}, \lambda_q)\, I\big(\theta_e(\mathbf{x}, \lambda_q) + \mathbf{u}, \lambda_q\big) \;+\; B_e(\mathbf{x}) \;+\; n_e(\mathbf{x}),$$

$$n_e \sim \text{Huber}(0,\ \sigma_e), \qquad \sigma_e^2 = \big(\nu_{d,f}\, \sigma_e^{\text{pipe}}\big)^2 + \big(f\, S_e\big)^2.$$

4.1 Pointing and distortion

$$\theta_e(\mathbf{x}, \lambda) = (I + M_e)\big[\theta^{\text{wcs}}(\mathbf{x}) + \delta_e + \Delta(\mathbf{x}; \lambda, t_e) - \mathbf{c}_e\big] + \mathbf{c}_e + \mathbf{t}_e.$$

4.2 Kernel shape

$$k_e(\mathbf{u}; \mathbf{x}, \lambda) = \frac{\exp \ell_\vartheta\big(R_e^\top \mathbf{u};\ h_e(\mathbf{x}, \lambda)\big)}{\sum_i w_i \exp \ell_\vartheta\big(R_e^\top \mathbf{u}_i;\ h_e\big)}.$$

The conditioning code is a sum of parts, each scaled by its own variability scale:

$$\begin{aligned} h_e(\mathbf{x}, \lambda) = \bar{h}(\lambda_{\text{piv}}, \mathbf{T}_f, \text{inst}) + \tau_\lambda\big[\bar{h}(\lambda) - \bar{h}(\lambda_{\text{piv}})\big] + \tau_{\text{det}}\, \hat{h}(d, \tilde{\mathbf{x}}) \\ + \tau_{\text{ep}}\, \hat{h}(t_e) + \tau_{\text{exp}}\, \mathbf{z}_e, \qquad \mathbf{z}_e \sim \mathcal{N}(0, I). \end{aligned}$$

- ℓ_ϑ is an MLP with FiLM conditioning.
- Its input is a log-radius and angular-harmonic encoding of the offset, out to 10″.
- R_e rotates into the detector frame.

4.3 Throughput

$$\begin{aligned} g_e(\mathbf{x}, \lambda) = \frac{T_f(\lambda)}{\lambda N_f} \exp\Big[\bar{\gamma}(\lambda, \mathbf{T}_f, \text{inst}) + \tau_{gd}\, \hat{\gamma}(d, \tilde{\mathbf{x}}, \lambda) \\ + \tau_{ge}\, \hat{\gamma}(t_e, \lambda) + \tau_{gx}\, u_e - m(\lambda)\Big]. \end{aligned}$$

4.4 Background

$$B_e(\mathbf{x}) = \sum_{i+j \le 2} b_{e,ij}\, P_i(\tilde{x}_1) P_j(\tilde{x}_2) + \Pi_d(\tilde{\mathbf{x}}) + r_e(\text{row}) + c_e(\text{col}).$$

4.5 Rendering, variability scales and gauges

- **Rendering:** overlap-add FFT, checked against a per-pixel quadrature to 6×10^{-4}–2×10^{-3}.
- **Variability scales** τ:
  - random effects by a type-II (EM) update;
  - function axes by held-out validation evidence with an off-switch check.
- **Gauges:**
  - **flux:** exact, via $m(\lambda)$;
  - **blur:** a floor;
  - **shift:** zero-mean translations;
  - **pedestal:** training background constants sum to zero about each exposure's median.

5. Evaluation and current results

$$\chi^2/N = \frac{1}{|\mathcal{M}|} \sum_{(e,\mathbf{x})\in\mathcal{M}} \left(\frac{d_e(\mathbf{x}) - \hat{d}_e(\mathbf{x}) - b_e}{\nu_d\, \sigma_e^{\text{pipe}}(\mathbf{x})} \right)^2, \qquad \mathcal{M} = \{\text{held-out pixels with S/N}_{\text{ref}} \geq 5\}.$$

- **Held-out pixels:** these are held-out dither exposures. Only per-exposure nuisances may be refitted on them.
- **Masks:** the masks and S/N_{ref} come from the official stage-3 mosaic.

Field	fieldop	Official JWST pipeline (mosaic blotted back)	Empty sky
SMACS NIRCam, 20″ box	1.77 [1.68, 1.86] (older package); 3.29 (newer, regressed)	1.80 [1.61, 2.02]	126
LMC NIRCam, 10 patches	8.54 [7.32, 9.90]	6.14 [5.25, 7.22]	30.3

The field operator is better than the pipeline at S/N < 20 and worse at S/N > 50, the bright star cores.

6. Defects found in review

1. **Confirmed: blur-floor gradient.** The dilation's gradient is wrong: one pass runs without autograd. Kernel gradients are biased by 8–26%, and the sign is wrong at large dilation.
2. **Design defect: the throughput field absorbs gains.** It also absorbs per-detector gain constants, so it is always kept by the variability rule. This is the likely cause of the SMACS regression.
3. **Design defect: the pedestal gauge.** It is pinned at exposure medians, which conflicts with sky positivity in crowded fields.
4. **Minor:**
   - the `--tau-prior` flag is ignored;
   - the logged batch χ^2 is not importance-weighted;
   - there is slight validation leakage in registration;
   - the WCS source for `_rate` input is wrong;
   - inf can turn into NaN.

Geometry, orientation, flux, held-out isolation and the renderer were checked numerically and are correct.

7. Questions for the choice of functional forms

1. **Which approximations to keep or drop.** Which steps of the ladder should be dropped because keeping the more general form *breaks* degeneracies? Candidates:
   - foreground through the response (A4: a sky flat);
   - focal-plane continuity of the optics (A3);
   - model-based Poisson variance (A7 L1);
   - an explicit inter-pixel capacitance / brighter-fatter stage (A5).
2. **Sky** $I(\theta, \lambda)$. Which continuous representation keeps point sources sharp below the finest pixel, represents extended emission compactly, optimises as well as a grid and scales to the full sky? Is a sparse continuous component acceptable without a catalogue?
3. **Blur gauge for a continuous sky.** Band-limit the sky at the sharpest kernel, fix a reference kernel, or only declare and report it?

4. **Kernel family.** One conditional field, or the composition $k^{\mathrm{opt}}(\mathbf{p}) * k_d^{\mathrm{det}}(\mathbf{x})$, each a field? Which conditioning (FiLM, hypernetwork, additive codes) gives smooth, identifiable variation across $\mathbf{p}$, λ and t?
5. **Geometry and the kernel.** Keep the distortion separate, or let it be the kernel's first moment, with the internal gauge pinned?
6. **Throughput.** Should it factor into filter $T_f(\lambda; \mathbf{p})$ × quantum efficiency/flat $\eta_d(\mathbf{x}, \lambda)$ × gain g_e, so that per-detector gains are separated from spatial structure?
7. **Foreground.** What form for I_{fg} (smooth in θ and λ, varying with epoch and pointing) breaks the sky–foreground gauge without a physical zodiacal-light model?
8. **Variability scales.** What principled evidence decides whether an axis is kept: a Laplace or marginal likelihood on a linearised operator, refitted likelihood ratios, or cross-validation? How can weak but real effects (the PSF's colour dependence within a filter) be kept?
9. **Noise.** Should the model work at L1 with model-based variance, with correlated noise as covariance or as nuisances? How should outliers be handled without a mixture? An earlier mixture classified bright cores as outliers and failed.
10. **Cost.** About 1.2 s per step on an 11 GB GPU slice, dominated by fine-level FFTs. Which forms make the operator cheap at survey scale?